

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

20MEE93 - INTRODUCTION TO BUSINESS ANALYTICS WITH PYTHON

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) **SQC** Tables are permitted.

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | | | | |
|---|---------|---------|---------|-----|-----|----|---|-----|----|---|---|----|---|----|---|---|-----|-----|---|----|---|---|
| 1. a) Briefly explain the different types and levels of measurement of the data. | 6 | L*2 | 1 | 1 | | | | | | | | | | | | | | | | | | |
| b) Discs stamped out of sheet metal are found to have weight variation confirming to normal distribution, with mean weight of 3.827 kg and standard deviation of weight 0.03 kg. The specification weight for these discs is 3.83 ± 0.05 kg. Determine the percentage of discs which are expected not to meet the specification. | 4 | L3 | 1 | 2 | | | | | | | | | | | | | | | | | | |
| c) Write a program input the following table using PANDAS. Also add the python/ PANDAS codes to add the items in "Series1" and "Series2" and present it in the column 4. | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1"> <thead> <tr> <th>Index</th> <th>Series1</th> <th>Series2</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>100</td> <td>20</td> </tr> <tr> <td>B</td> <td>-20</td> <td>80</td> </tr> <tr> <td>C</td> <td>-</td> <td>30</td> </tr> <tr> <td>D</td> <td>50</td> <td>-</td> </tr> <tr> <td>E</td> <td>-60</td> <td>-20</td> </tr> </tbody> </table> | Index | Series1 | Series2 | A | 100 | 20 | B | -20 | 80 | C | - | 30 | D | 50 | - | E | -60 | -20 | 5 | L3 | 2 | 2 |
| Index | Series1 | Series2 | | | | | | | | | | | | | | | | | | | | |
| A | 100 | 20 | | | | | | | | | | | | | | | | | | | | |
| B | -20 | 80 | | | | | | | | | | | | | | | | | | | | |
| C | - | 30 | | | | | | | | | | | | | | | | | | | | |
| D | 50 | - | | | | | | | | | | | | | | | | | | | | |
| E | -60 | -20 | | | | | | | | | | | | | | | | | | | | |
| d) The finance department of a company wants to calculate the monthly pay of one of its employee. Monthly pay should be calculated as mentioned in the formula below and display all the employee details. Monthly Pay= No. of hours worked in a week * Pay rate per hour * No. of weeks in a Month. Write a Python Program to implement the problem. | 5 | L3 | 2 | 2 | | | | | | | | | | | | | | | | | | |
| 2. a) Control charts on $\bar{X}$ and R are to be maintained for samples of size 5 are to be maintained on the tensile strength in pounds of yarn. 30 samples were selected and the means and ranges of each sample were computed. This yielded $\sum \bar{X} = 607.8$ and $\sum R = 144$. | | | | | | | | | | | | | | | | | | | | | | |
| i. Find the control limits for $\bar{X}$ and R charts. | | | | | | | | | | | | | | | | | | | | | | |
| ii. Find the process capability | | | | | | | | | | | | | | | | | | | | | | |
| Assume that both $\bar{X}$ and R charts exhibit control. There is a single lower specification limit of 16lbs. If the tensile strength is normally distributed, what fraction of yarn would fail to meet the specification? | 6 | L3 | 1 | 2 | | | | | | | | | | | | | | | | | | |
| b) What do you mean by quality? Explain quality engineering. | 4 | L3 | 1 | 2 | | | | | | | | | | | | | | | | | | |
| c) Write a Python program using 'def' function and 'if' statement to compare whether the given/ input number is greater than, equal to or less than 10 and print the result as statements. | 6 | L2 | 3 | 1 | | | | | | | | | | | | | | | | | | |
| d) Write the syntax of range function in python and illustrate its working with an example. | 4 | L2 | 2 | 1 | | | | | | | | | | | | | | | | | | |

3. a) The general manager of a building material production plant feels that the demand for plasterboard shipments may be related to the number of construction permits issued in the country during the previous quarter.

Construction permits	15	9	40	20	25	25	15	35
Plasterboard Shipments	6	4	16	6	13	9	10	16

- i) Compute the values of slope b and intercept a.
 ii) Determine a point estimate for the plasterboard shipment when the number of construction permits is 30.
 iii) Determine the coefficient of correlation.
- b) Consider a list A=["Hello", 12, 15, 20.56]. Write the output of the following.
 i) A[1:3]
 ii) A[-2]
 iii) A[-1:-3]
 iv) A[-1:]
 v) A[1:]
 vi) A.append("World")
- c) State any 2 differences between
 i) numpy and pandas
 ii) lists and tuples

10 L3 1 1

6 L3 1 2

4 L3 2 2

Unit – II

4. a) Consider the following model:
 Minimize $Z = 40x_1 + 50x_2$,
 Subject to
 $2x_1 + 3x_2 \geq 12$
 $x_1 + x_2 \geq 12$
 $2x_1 + x_2 \geq 20$
 and $x_1, x_2 \geq 0$
 i) Use the graphical method to solve this model.
 ii) How does the optimal solution change if the objective function is changed to $Z = 40x_1 + 70x_2$?
- b) Use the concept of percentage point function and cumulative distribution function to **write a program using python** to solve the following problem. Import the required modules, show the necessary comments and outputs wherever necessary.
A school claimed that the students' study there is more intelligent than the average school. On calculating the IQ scores of 50 students, the average turns out to be 110. The mean of the population IQ is 100 and the standard deviation is 15. State whether the claim of principal is right or not at a 5% significance level.
- c) **Write a python program** to solve the following Linear programming problem.
 Maximize $Z = 3000x_1 + 2000x_2$,
 Subject to
 $x_1 + 2x_2 \leq 6$
 $2x_1 + x_2 \leq 8$
 $x_2 \leq 2$
 $x_2 - x_1 \leq 1$
 $x_1, x_2 \geq 0$

10 L3 3 2

5 L3 4 2

5 L3 4 2

5. a) A manufacturer of packing material produces two types of packing tins – round and flat. Major production facilities involved are cutting and joining. The cutting department can process 300 tins of round or 500 tins of flat per hour. The joining department can process 400 tins of round or 300 tins of flat per hour. If the profit contribution of round tins is Rs. 2 per tin and that of flat is Rs. 5 per tin, formulate the problem as LPP and solve it graphically.
- b) **Write a python program** to solve the following control chart problem and draw the $\bar{X}$ and R control charts. The data given below represents percentage water absorption capacity of a common building brick. 25 samples of 4 bricks were taken.

Sample No.	$\bar{X}$	R	Sample No.	$\bar{X}$	R
1	15.01	9.1	14	9.8	17.5
2	12.3	9.9	15	8.8	10.5
3	7.4	9.7	16	8.1	4.4
4	8.7	6.7	17	6.3	4.1
5	8.8	7.1	18	10.5	5.7
6	11.7	9.1	19	9.7	6.4
7	10.2	12.1	20	11.7	4.6
8	11.5	10.8	21	13.2	7.2
9	11.2	13.5	22	12.5	8.3
10	10.2	6.9	23	7.5	6.4
11	9.6	5	24	8.8	6.9
12	7.6	8.2	25	8	6.4
13	7.6	5.4			

6. a) Solve the following LPP by graphical method:

$$\text{Maximize } Z = 2.8x_1 + 2.2x_2$$

Subject to,

$$x_1 \leq 20,000$$

$$x_2 \leq 40,000$$

$$0.003x_1 + 0.001x_2 \leq 66$$

$$x_1 + x_2 \leq 45,000$$

$$x_1, x_2 \geq 0$$

- b) Briefly explain the assumptions related to Linear programming problems.
- c) **Write a python program** to solve the following linear regression and correlation problem. Import the required modules, show the necessary comments and outputs wherever necessary.

Month	1	2	3	4	5
Average temperature	24	41	32	30	38
Resort attendance('000s)	43	31	39	38	35

The following table gives the 5 months of average monthly temperature and corresponding monthly attendance. Compute linear regression equation of the relationship between the two if next month's average temperature is forecast to be 45 degrees.

- i) Use linear regression equation to develop the forecast.
- ii) Compute correlation coefficient for the above data and determine the strength of linear relationship between average temperature and attendance.
- iii) Plot the graph of Average temperature vs. resort attendance

Unit – III

7. a) A firm owns facilities at seven places. It has manufacturing plants at places A, B and C with daily output of 500, 300 and 200 units of an item respectively. It has warehouses at places P, Q, R and S with daily requirements of 180, 150, 350 and 320 units respectively. Per unit shipping charges on different routes are given below.

To:	P	Q	R	S
From A:	12	10	12	13
From B:	7	11	8	14
From C:	6	16	11	7

The firm wants to send the output from various plants to warehouses involving minimum transportation cost. How should it route the product so as to achieve its objective?

- b) Solve the following transportation problem to maximize the profit.

Destination	P	Q	R	S	Supply
Source					
A	6	11	9	8	200
B	13	10	8	15	300
C	9	9	14	12	500
D	12	10	12	10	100
Demand	350	250	300	200	

8. a) Solve the following transportation problem for optimum transportation cost.

		To Market			Supply
		A	B	C	
From Warehouse	1	10	12	7	180
	2	14	11	6	100
	3	9	5	13	160
	4	11	7	9	120
Demand		240	200	220	

- b) A bakery produces a specially cake in either of its two plants, as follows:

Plant	Daily Production capacity, (cakes)	Production cost, (Rs./cake)
A	2500	23
B	2100	24

Four retailers are willing to purchase their cakes; their demands and prices they are willing to pay are as follows.

Retailer	Maximum Demand, (cakes)	Price offered (Rs./cake)
1	1800	39
2	2300	37
3	550	40
4	1750	36

The cost (in rupees) of packing and shipping a cake from a plant to a retailer is given in the following table:

	Retailer 1	Retailer 2	Retailer 3	Retailer 4
Plant A	6	8	11	9
Plant B	12	6	8	5

Solve this as a transportation problem and determine a delivery schedule for the bakery that will maximize its total profit from this cake.